# Hidden Dynasties: Dynasty Trusts and the Concentration of Wealth

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**Abstract**

Dynasty trusts let wealthy families pass assets across generations outside the estate tax while binding heirs to the principal. We quantify their effect on US wealth concentration with a calibrated overlapping-generations model in which founding, funding and drawing on a trust are household choices. In general equilibrium, trusts raise the top 1% wealth share by 3.2 percentage points and the top 0.1% share by 3.1 points; calibrations that fit the data almost as well give 2.4 to 3.6 points. Trusts work by adding to saving, and four-fifths of the effect comes from the combination of the estate-tax shield and lock-in, so taxing trust wealth once a generation would remove most of it in the long run.

Much of this concentration is invisible in official statistics. The Federal Reserve's Distributional Financial Accounts count trust assets but, lacking data on who benefits, spread them across all holders of similar assets. Because the model knows every trust's beneficiary, we can measure its wealth both ways. The official measure captures only a fifth to a half of the effect, and calibrating to it implies that the top 1% hold about 34.5% of US wealth, not the measured 32.5%.

**AI disclosure.** The model code, calibration and experiment scripts were written and run with the assistance of Claude, an AI model made by Anthropic, working under the author's direction. Claude also assisted in drafting and editing the text, and drafts were reviewed by separate AI model sessions. All modelling choices were approved by the author, who is responsible for the results (Appendix A.7).

**JEL codes:** D31, E21, H24, H21, D64. **Keywords:** dynasty trusts, estate tax, wealth inequality, top wealth shares, Distributional Financial Accounts, overlapping generations.

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## 1. Introduction

The US estate tax reaches about one death in 700. Yet the wealth it is meant to tax increasingly passes through vehicles it never sees. Since the mid-1990s, following South Dakota in 1983, about half the states have abolished or effectively abolished the rule against perpetuities, allowing trusts that last indefinitely. Assets placed in such a trust within the federal generation-skipping transfer (GST) exemption pass from generation to generation without estate or GST tax. That exemption is now $15 million per person, indexed and with no sunset, under the 2025 tax act; a couple can shelter $30 million if each spouse allocates their own, since unlike the estate-tax exemption the GST exemption is not portable. Sitkoff and Schanzenbach (2005) estimate that roughly $100 billion moved into perpetual-trust states within a decade of the first repeals. Recent estimates put assets in perpetual or non-grantor trusts at $4.5 trillion or more (Galle, Gamage and Lord 2026) and $5.4–10.8 trillion

(Congressional Research Service 2026). Galle, Gamage and Lord estimate that 80–90% of the wealth rich families set aside for heirs will never face the estate tax.

Avoidance of this kind is not new. The federal estate tax dates from 1916 and the gift tax, in lasting form, from 1932; before then, wealth passed between generations essentially untaxed. Since then the very wealthy have always had some way to pass wealth with control across generations, and what has changed is its cost. Until 1976, a trust lasting as long as the rule against perpetuities allowed could skip estate tax at every intermediate generation with no limit on its size, while wealth passed outright faced top estate-tax rates of up to 77%. The 1986 generation-skipping tax capped how much a trust can shelter, though trusts already irrevocable in 1985 were grandfathered, and the later repeal of perpetuity rules removed the time limit. Cooper (1979) called the estate tax a voluntary tax for this reason. What is new is the combination of a large, rising exemption with no limit on duration, together with ever wider access to the trust companies, states and instruments that make such trusts routine. This paper measures the long-run consequence of that combination.

This paper asks how much dynasty trusts raise the concentration of wealth at the top. The question is hard to answer from data alone, for two reasons. First, trust wealth is hard to see: it never passes through an estate, grantor trusts report on the founder's income tax return, and household surveys largely miss assets held for beneficiaries. Second, the answer is a counterfactual. Removing trusts would change how much families save, what they bequeath and how much they give to charity, and through capital supply it would change the interest rate.

We therefore build a quantitative overlapping-generations model in which founding a trust is a choice. Households face income and entrepreneurial-return risk, value bequests and charity, make lifetime gifts and pay estate and gift taxes. In any period they can found a dynasty trust, which shields assets from estate tax at every generation but locks the principal away from heirs. We calibrate the model to US top wealth shares, estate-tax and charitable-bequest data, the share of transfers made during life, and outside estimates of the share of top wealth held in trusts. We then remove trusts, re-solve the interest rate, and compare stationary equilibria.

A central step is measurement. The Federal Reserve's Distributional Financial Accounts (DFA), the benchmark for US top wealth shares, distribute Financial Accounts totals that include personal trusts. But they allocate those totals using the Survey of Consumer Finances, which records few trusts. Trust wealth is therefore counted but mostly spread across all holders of similar assets, not attributed to the families who benefit. We measure model wealth the same way and calibrate to it.

We have four main findings.

1. **Dynasty trusts raise top concentration materially.** In our main calibration, where trusts hold 19% of top-1% wealth (near the top of the CRS range of 10–20%), they raise the top 1% wealth share by 3.2 percentage points (34.5% vs 31.3%) and the top 0.1% share by 3.1 points. Calibrations that fit almost as well give 2.4 to 3.6 points.
2. **Much of the effect is invisible in official statistics.** Measured the DFA way, the top-1% effect is 1.3 points, and because we calibrate to the DFA-measured share, the model implies a true top-1% share about 2 points higher (34.5% against 32.6%). At the DFA's own spreading rule, between 45% and 80% of the effect is hidden, depending on how much trust wealth the SCF captures; how the DFA spreads the rest matters even more.

3. **The effect comes from the shield and lock-in together.** Either feature alone gives at most a sixth of the effect, even when heirs can withdraw only part of the principal; their interaction gives four-fifths. Trusts with a tax shield but no lock-in are drawn down by heirs; trusts with lock-in but no shield only relabel wealth.
4. **Trusts add to saving rather than relabel it.** Without trusts, capital is 6.0% lower at the calibrated interest rate, and the equilibrium rate rises from 5.99% to 6.21%.

The size of the effect depends on how much top wealth sits in trusts, which is poorly measured. In the alternative calibration, re-solving the model to hit trust shares of top-1% wealth from 10% to 27% gives effects from 0.8 to 4.1 points, rising faster than the trust share. It also depends on who holds trust wealth. The more founders value trusts beyond their tax advantages, the more merely rich families use them, and the smaller the effect of a given amount of trust wealth: the calibration with $\theta = 1.38$ gives 2.4 points. In the alternative calibration, treating lifetime gifts more strictly (+2.9 points) or giving trust assets a higher return (+2.5 points) leaves the effect at or above its baseline of 2.4. In the main calibration, denying the shield to trust funding above the GST exemption lowers the effect to +1.4 points, mainly by lowering trust wealth to 11% of top-1% wealth.

The paper makes three contributions. To our knowledge, it gives the first general-equilibrium estimate of the effect of dynasty trusts on the wealth distribution in which trust founding, funding and payouts are household choices. It shows how to measure a model economy the way the DFA measures the US, which matters most for what official statistics would show; the measurement choice itself moves the true effect little once the founder's valuation of trusts is held fixed (Sections 3.4 and 5.2). And it isolates the interaction of tax shield and lock-in as the mechanism, which maps directly onto the two policy levers, duration limits and periodic taxation. Along the way, it maps the routes by which US wealth actually passes between generations, from annual-exclusion gifts and grantor trusts to valuation discounts and the step-up in basis, and shows which of them a quantitative model represents and which it leaves out (Table 1). The calibration targets, experiments and predictions were recorded in advance in a dated project log, and the code is public.

Section 2 describes dynasty trusts and related work. Section 3 sets out the model, calibration and measurement. Section 4 presents the results, and Section 5 discusses policy and limitations. Section 6 concludes.

## 2. Background and related literature

### 2.1 Dynasty trusts

A dynasty trust is an irrevocable trust designed to hold assets for many generations. Two features of US law make it attractive.

- **Transfer-tax shield.** Since 1986, transfers that skip a generation face the GST tax, but each person has a GST exemption. Assets placed in a trust within that exemption, and all their later growth, pass to grandchildren and beyond with no estate or GST tax. The exemption is $15 million per person from 2026, indexed and permanent under the 2025 tax act.
- **Duration.** The common-law rule against perpetuities limited how long a trust could last to lives in being plus 21 years, which in practice allows trusts of about 80 to 100 years or more. South Dakota abolished the rule in 1983, and starting in the mid-1990s about half the states abolished it or extended permitted durations to hundreds of years. Sitkoff and Schanzenbach (2005) show that abolition attracted trust

business: by 2003, abolishing states gained about $6 billion in trust assets each on average, about $100 billion in total, but only if they did not tax trust income from out-of-state settlors.

A third feature, lock-in, comes from the trust instrument. Typical dynasty trusts restrict distributions of principal and include spendthrift clauses, so beneficiaries receive income or discretionary payments rather than control of the capital. This protects the assets from beneficiaries' creditors and spending. It is also what keeps the wealth intact across generations.

**What we mean by a dynasty trust.** We define the term by function, not legal form: a dynasty trust is any vehicle that keeps assets outside the transfer-tax base across generations (the shield) and restricts beneficiaries' access to principal (lock-in). Many instruments meet this definition, including GST-exempt long-term trusts, intentionally defective grantor trusts with GST exemption allocated, and insurance trusts structured the same way. Instruments with only one of the two features fall outside it: revocable trusts and marital (QTIP) trusts are taxed in an estate, charitable trusts move wealth out of the family, and trusts with broad withdrawal rights have no lock-in. Outside estimates of trust wealth do not all use this definition, which we return to in Section 3.4.

**Funding beyond the exemption.** The exemption caps what can be placed in a trust free of gift tax, but planning techniques move further value in at little tax cost. In a sale to an intentionally defective grantor trust, the founder sells appreciating assets to the trust for a note, so growth above the note's interest rate accrues to the trust, and the founder's payment of the trust's income tax is a further untaxed transfer. Grantor retained annuity trusts (GRATs) shift growth above the IRS hurdle rate in the same way, though their remainders are harder to place in a GST-exempt trust. Valuation discounts on family partnership interests reduce the taxable value of what is transferred. Funding a trust above the GST exemption is costly for a second reason: the trust then has an inclusion ratio above zero, and that share of every later distribution to grandchildren or termination in their favour pays GST tax at 40%. Planners therefore keep dynasty trusts within the exemption and grow them with the techniques above. Our model does not represent these techniques one by one. Instead, the baseline lets a family fund a trust beyond the exemption by paying gift tax on the excess, and then shields the whole trust. This is a stand-in for the freeze route: it charges more at funding than a freeze would, but it spares the excess the GST tax that direct funding above the exemption would bear. Because the calibration matches the amount of wealth held in trusts, the route matters less than the amount. Two checks bracket it: denying the shield to any funding above the exemption (Section 4.5), and adding explicit freeze transfers into the trust (Section 5.2).

### 2.2 How wealth passes between generations

Dynasty trusts are one of about a dozen routes by which wealth passes to the next generation, and each route is taxed differently in three ways: the transfer tax, the income-tax basis heirs inherit, and how freely heirs can use what they receive. Quantitative models of wealth inequality usually reduce all of this to a single estate tax on bequests. Table 1 sets out the main routes under 2026 law and how our model treats each.

**Table 1. Routes for passing wealth to heirs (US federal law, 2026)**

| Route | How it works | Transfer tax | Heirs' tax basis | In our model |
|---|---|---|---|---|
| Bequest at death | Assets pass through the estate | 40% above the exemption: $15M per person, $30M per couple using portability (IRC §2010) | Stepped up to market value at death (§1014) | Yes; $30M per family |
| Annual-exclusion gifts | $19,000 per donor per recipient per year; tuition and medical bills paid directly are unlimited (§2503(b), (e)); gifts to a trust qualify only if beneficiaries get temporary withdrawal (Crummey) rights, and for GST purposes only under the stricter rules of §2642(c) | None; no exemption used | Carryover: the giver's basis (§1015) | Approximated: $300k free once, at the parent's age 55 |
| 529 front-loading | Five years of exclusions at once: $95,000 per donor per child (§529(c)(2)(B)) | None | Growth tax-free if spent on education | Inside the $300k |
| Gifts above the exclusion | Use the unified lifetime exemption, then 40% | Exemption, then 40% | Carryover | Yes; exemption not deducted later (approximation; Section 3.2) |
| Dynasty (GST-exempt) trust | Funded within the GST exemption (§2631); perpetual in about half the states | None, at any generation | Carryover; no step-up at later deaths | Yes: the paper's subject |

| Route | How it works | Transfer tax | Heirs' tax basis | In our model |
|---|---|---|---|---|
| Grantor trust and sale to it | Founder pays the trust's income tax (not a gift: Rev. Rul. 2004-64); sells appreciating assets to the trust for a note | Growth above the note's rate passes free | Carryover; no step-up at the founder's death (Rev. Rul. 2023-2), but low-basis assets can be swapped back into the estate | Grantor tax yes; sales only in the freeze check (Section 5.2) |
| GRAT | Annuity back to the founder; growth above the §7520 rate goes to heirs (§2702) | Close to zero if "zeroed out" | Carryover | Freeze check only |
| Valuation discounts | Family partnership or LLC interests valued below their share of the assets (lack of control, marketability) | Lowers the taxable value of gifts and estates | Carryover or step-up at the discounted value | Coded, switched off |
| Life-insurance trust | Trust owns a policy on the founder's life | Proceeds outside the estate | Proceeds not income-taxed | No |
| Charitable vehicles | Private foundations; lead trusts pay charity first, then heirs | Deductible; a lead trust's remainder can pass at low gift value | — | Charity at death only |
| Transfers to a spouse | Unlimited marital deduction (§2056); unused exemption is portable | None between spouses | Step-up at the first death for the decedent's share | No spouses; the $30M family exemption stands in |
| Buy, borrow, die | Borrow against appreciated assets instead of selling them | Estate tax at death as usual | Step-up erases the gain | No: capital income taxed as it accrues |

*Sources: IRS 2026 amounts as reported by Morgan Lewis (2025); Internal Revenue Code sections and revenue rulings as cited.*

Three features of this landscape matter for our question. First, the routes differ in how much control they leave heirs, and only trusts combine a permanent transfer-tax shield with restricted access. Second, several routes enlarge the effective exemption: annual exclusions used every year and across grandchildren, valuation

discounts, and freeze transfers. Our model represents these only through its calibration to the effective estate-tax rate on large estates, which pins down total avoidance but not its mix across routes. Third, the transfer tax and the income tax interact through basis: inheriting through an estate erases capital gains, while gifts and trusts carry them forward. Section 6 discusses what modelling basis would add.

The scale of trust wealth is uncertain. Galle, Gamage and Lord (2026) estimate at least $4.5 trillion in perpetual trusts, one-third or more of the wealth of the top 0.1%. The Congressional Research Service (2026) estimates $5.4–10.8 trillion in non-grantor trusts from fiduciary income tax returns. South Dakota alone reports about $815 billion in trust assets at the end of 2024 (Strubinger 2025).

### 2.3 Measuring top wealth

US top wealth shares come from three main sources, each of which sees trusts poorly. Estate tax returns (Kopczuk and Saez 2004) never see dynasty-trust assets, because the assets never pass through an estate. Capitalised income (Saez and Zucman 2016; Smith, Zidar and Zwick 2023) misses assets that generate little taxable income and attributes grantor-trust income to the grantor. The SCF and the DFA built on it (Batty et al. 2019) record trust assets only partly. Our measurement exercise (Section 3.5) shows what this implies for measured top shares in a model where trust wealth is known.

### 2.4 Quantitative models of bequests and wealth concentration

Standard heterogeneous-agent models with uninsurable earnings risk (Aiyagari 1994; Huggett 1996) produce realistic wealth dispersion but too thin a top tail. The literature has added four mechanisms to thicken it: extreme earnings states (Castañeda, Díaz-Giménez and Ríos-Rull 2003), bequest motives (De Nardi 2004), entrepreneurship (Quadrini 2000; Cagetti and De Nardi 2006) and return heterogeneity (Benhabib, Bisin and Zhu 2011; Benhabib, Bisin and Luo 2019; Gabaix et al. 2016); De Nardi and Fella (2017) survey them. Our model combines the first three with a reduced form of the fourth. We model bequests as a warm glow rather than through altruism toward heirs' utility (Barro 1974; Becker and Tomes 1979), which keeps the model tractable with a two-generation state and lets bequests be a luxury good, as the data require.

We discipline return heterogeneity rather than assume it. The evidence agrees that the rich earn higher returns but attributes most of the gap to risk, scale and owners' human capital rather than a free premium (Moskowitz and Vissing-Jørgensen 2002; Fagereng et al. 2020; Smith, Yagan, Zidar and Zwick 2019); consistent with this, most large fortunes today start as founders' businesses rather than inheritances (Kaplan and Rauh 2013). We therefore cap the entrepreneurial premium and let high-growth owners earn superstar labour income. For the same reason, trust assets earn the family's own return in our baseline. Earlier work on family offices and dynasty trusts posits a return advantage for professionally managed dynastic wealth; we treat it as a sensitivity (Section 4.5) because it would build the result into the assumptions.

Studies of estate taxation in these models, such as Cagetti and De Nardi (2009) and De Nardi and Yang (2016), tax bequests at statutory rates and do not model the trust route around the tax.

A related empirical literature studies avoidance of the estate tax: Kopczuk (2007) on late-life tax planning, and Raub, Johnson and Newcomb (2010) on the gap between Forbes 400 wealth and the wealth reported on their estate tax returns. Our results quantify one channel of that avoidance in general equilibrium.

Our measurement result connects to the literature on hidden wealth. Zucman (2013) and Alstadsæter, Johannesen and Zucman (2019) show that offshore holdings, concentrated at the very top, make measured top shares understate concentration. Dynasty trusts are a domestic and legal counterpart: counted in national totals, but not attributed to the families who benefit. Whose wealth a discretionary trust is also sits at the centre of wealth-tax design (Saez and Zucman 2019). On the normative side, work on optimal bequest taxation (Farhi and Werning 2010; Piketty and Saez 2013) takes the tax base as given; our results describe how a legal vehicle erodes that base, a margin those formulas would need to include.

## 3. Methodology

We measure the effect of dynasty trusts on top wealth concentration by building a quantitative overlapping-generations (OLG) economy in which trusts are a choice, then removing them and comparing stationary equilibria. The model earns its answer: nothing about the size of the trust effect is assumed. Households decide whether to found a trust, how much to put in it and what to bequeath, and the trust share of top wealth is a calibration target, not an input.

The economy has four building blocks, added in layers so each can be tested against the one below it:

1. **Life cycle and bequests (Layer A).** Households work, save, retire and die with a warm-glow bequest and charity motive, facing income, capital and estate taxes.
2. **Fortune creation (Layer B).** A reduced-form entrepreneurial return process generates the thick right tail of wealth.
3. **Wealth-preservation technology (Layer C).** Households can found a dynasty trust, which shields assets from estate tax across generations but locks principal away from heirs. Parents can also make lifetime gifts.
4. **General equilibrium.** A corporate sector sets the interest rate and wage; in every experiment the interest rate is re-solved so the capital market clears.

Each layer nests the one below it: switching trusts off reproduces Layer B exactly, and switching off the entrepreneurial process reproduces Layer A. These nesting checks are part of the test suite.

### 3.1 Households

**Demographics.** A model period is five years. Households enter at age 20 and live at most 14 periods, to age 89, retiring after nine periods (age 65). Survival probabilities follow the US period life table, except that nobody dies before age 60, so every parent is alive when the child forms a household. Each household has children one generation (seven periods, 35 years) after entry, and the population grows at 1% a year. Because a generation is seven periods and life is fourteen, every household overlaps with its parent for part of life; the model tracks the joint distribution of parent and child, which is what lets a parent's trust, gift or estate land on a child whose own state is known.

**Preferences.** Households maximise expected lifetime utility with CRRA period utility ($\sigma = 1.5$) and an annual discount factor $\beta$. At death, the estate is split between a warm-glow bequest and charity:

$$B(b,g) = \phi_1 \left[ \left(1 + \frac{b}{\phi_2}\right)^{1-\sigma} - 1 \right] + \chi \left[ 1 - \left(1 + \frac{g}{\phi_{2c}}\right)^{1-\sigma} \right] \tag{1}$$

where b is the net bequest after estate tax and g is charitable giving. With $\sigma > 1$, a negative $\varphi_1$ and a positive $\chi$ make both terms increasing in what is given. The shifters $\varphi_2$ and $\varphi_2$c make bequests and charity luxury goods (De Nardi 2004). The bequest strength $\varphi_1$ and charity strength $\chi$ are calibrated; like $\varphi_2$ and $\varphi_2$c, they are stated in units where mean five-year earnings equal one, and utility is rescaled accordingly.

**Earnings.** Labour productivity follows a five-state persistent process ($\rho = 0.85$, variance 0.30 per period) on top of an age profile. A child's initial productivity is drawn conditional on the parent's, with persistence set to 0.545 so the model reproduces the earnings rank-rank slope of 0.341 in Chetty et al. (2014). Labour income is taxed at 30.1% (OECD *Taxing Wages*, 2026). Retirees receive a pension of 40% of mean earnings.

**Entrepreneurial returns.** The right tail of wealth comes from a reduced-form return process with three states: normal (N), entrepreneur (L) and high-growth entrepreneur (H). In states L and H, wealth earns a risky, right-skewed business return:

$$G = (1 + r + \pi_z)^5 \exp\left(s_z \xi - s_z^2/2\right), \qquad s_z = \sigma_z \sqrt{5} \tag{2}$$

with an annual premium of 1% in state L (fixed, following Moskowitz and Vissing-Jørgensen 2002) and close to zero in state H, and annual volatilities of 6.9% (L) and 61.7% (H). Following Smith, Yagan, Zidar and Zwick (2019), most of the return to high-growth owners is a return to their own human capital: while working, H owners earn 12.8 times the labour income of others with the same productivity. States move within a life: N→L with probability 6.9% and L→H 3.1% per period, with exit back to N at 20%. A child of an entrepreneur starts in L with probability 0.20 (0.07 otherwise). Capital income is taxed at an effective 18% as it accrues. The model does not track the tax basis of assets, so there are no unrealized gains, no step-up in basis at death and no carryover basis for gifts and trusts (Sections 5.3 and 6). These parameters come from the Layer B calibration (sum of squares 56.2 on its own targets) and are held fixed in the final calibration.

### 3.2 Transfers: bequests, lifetime gifts and dynasty trusts

Wealth crosses generations through three channels, and households choose among them.

**Bequests.** At death, the liquid estate above the exemption E = \$30M (the 2026 per-couple basic exclusion, rounded) is taxed at 40%. The household splits the estate between heirs and charity; charity is deductible. Estates and trusts are divided among children; with 1% annual population growth a household has about 1.42 children.

**Lifetime gifts.** When a child forms a household (parent aged 55), the parent makes one gift decision from a menu (1%, 3%, 8%, 20% or 40% of cash on hand). The first \$300k per family is free of gift tax. This stands in for the annual exclusion (\$19,000 per donor per recipient in 2026): two parents, about 1.4 children and five years give about \$270k, and tuition and medical bills paid directly are excluded without limit. In reality the exclusions can be used every year and for grandchildren and other relatives, so a family that gives systematically

over decades can pass far more than this tax-free. On this point the model is conservative. Larger gifts are taxed at 40% only beyond the exemption. As an approximation, the model does not deduct such gifts from the exemption available later, when funding a trust or at death, whereas the law does; this makes very large gifts too cheap. A bound in which gifts above the allowance get no exemption at all raises the trust effect in the alternative calibration from +2.4 to +2.9 points, as families substitute from gifts into trusts (Section 4.5). A separate warm-glow term (strength $\phi_g$, calibrated) values giving while alive.

**Dynasty trusts.** In every period, a household without a trust can found one. Founding:

- costs \$50k in legal fees, plus a planning friction $\kappa$ drawn each period from a logistic distribution with mean $\mu_f$ and scale $\nu_f$, as a share of wealth. Both are calibrated, so take-up rises smoothly with wealth rather than jumping at a threshold;
- moves an amount chosen from a menu (the remaining exemption, or 10–85% of cash on hand) into the trust. Funding is free of gift tax up to the exemption and taxed at 40% (tax-exclusive) above it. This taxed excess stands in for the freeze techniques, such as sales to grantor trusts and GRATs, that real families use to grow trusts beyond the exemption (Section 2.1).

Once founded, the trust has two properties, which the experiments switch off one at a time.

- **Tax shield.** Trust assets pass to heirs outside the taxable estate at every generation: no estate or generation-skipping tax.
- **Lock-in.** Heirs cannot reach principal. Each period they choose to retain the trust's income or have it paid out, or to add their own exemption to it. Winding up is not allowed (dead-hand terms and spendthrift clauses), except that trusts below \$100k are terminated as uneconomic (the Uniform Trust Code's default threshold, §414, is \$50,000; we use a higher one to reflect fixed costs).

Trust assets earn the family's return (same shock as its liquid wealth) and pay a trustee fee of \$10k a year plus 0.2% of assets. While the founder lives, it is a grantor trust and the founder pays its income tax; after that, the trust's income is taxed at 24%. The founder is not a beneficiary: a retained interest would pull the trust back into the estate (IRC §2036).

**How founders value a trust.** In the bequest motive, a dollar left in trust counts $\theta$ times a dollar bequeathed directly:

$$B(b,d,g) = \phi_1 \left[ \left( 1 + \frac{b + \theta d}{\phi_2} \right)^{1-\sigma} - 1 \right] + \chi \left[ 1 - \left( 1 + \frac{g}{\phi_{2c}} \right)^{1-\sigma} \right] \tag{3}$$

where d is trust assets. The same weight applies when an heir who holds a trust dies, so $\theta$ also shapes heirs' choices to keep income in the trust. $\theta$ is calibrated: $\theta < 1$ would mean founders discount money their heirs cannot touch, and $\theta > 1$ that they value the control and creditor protection a trust provides, or tax planning the model leaves out (Section 5.2). The main calibration gives $\theta = 1.04$; an alternative calibration with $\theta = 1.38$ fits almost as well (Section 3.4).

A second tier of advanced planning (valuation discounts, freeze transfers into the trust and lower-cost administration through a family office) is built into the code. It is switched off in the main results and used only in

one check in Section 5.2.

**The household's problem.** A household of age t enters the period with liquid wealth a, trust assets d, earnings state y, return state z and a role c: no trust, founder or heir. A return shock ξ is then realised and cash on hand is

$$m = \tilde{G}_z(\xi)\, a + e_t(y,z), \qquad \tilde{G}_z(\xi) = 1 + \big(G_z(\xi) - 1\big)(1 - \tau_a) \tag{4}$$

where $G_z$ is the five-year gross return of state z (the market return (1 + r)$^5$ in state N) and $e_t$ is after-tax earnings, (1 − $\tau_l$) w $\varepsilon_t e^y$ times the superstar multiplier in state H, or the pension after retirement. Trust assets earn the same return. An heir's trust is taxed as a separate entity; a founder's trust is a grantor trust, so it grows untaxed and the founder pays its tax:

$$D = \begin{cases} \max\{\, d\,[1 + (G_z(\xi) - 1)(1 - \tau_{ng})]\,(1-f)^5 - F,\ 0\,\} & \text{heir} \\ d\,G_z(\xi)\,(1-f)^5 - F, \quad m \leftarrow m - \tau_a\,(G_z(\xi) - 1)\,d & \text{founder} \end{cases} \tag{5}$$

where f = 0.2% is the proportional trustee fee and F the fixed fee over five years. (If the grantor tax exceeds half of cash on hand, the trust pays the rest. In practice such a reimbursement must be at the trustee's discretion, not the founder's right, or the trust is pulled back into the estate (Rev. Rul. 2004-64); the model treats it as automatic.)

The household then chooses among a small set of options o (Table 2), each defining a next-period role $c'_o$, trust balance $d'_o$ and cash left $m_o$:

**Table 2. Household options each period**

| Option | Who | Trust d′ | Cash left |
|---|---|---|---|
| Stay without a trust | no trust | 0 | m |
| Found and fund with x | no trust | x | m − K − x − $\tau_g$ max(x − E, 0) |
| Keep the trust as it is | founder, heir | D | m |
| Take this period's income | heir | $\ell$ min(d, D) | m + D − $\ell$ min(d, D) |
| Add x to the trust | founder, heir | D + x | m − x − $\tau_g$ max(x − $E_r$, 0) |
| Trust below $100k: wound up (forced) | heir | 0 | m + D |

Here K = $50k is the setup cost, x is either the remaining exemption or a share (10–85%) of cash on hand, $\tau_g$ = 40% is the gift tax, $E_r$ is the founder's remaining exemption max(E − d, 0) (an heir's own E), and $\ell$ is the lock-in share (1 in the baseline, 0 in the lock0 experiment). Given an option, the household chooses consumption and saving:

$$v_o = \max_{a' \geq 0}\ u(m_o - a') + \beta^5 \Big[\, s_t\, \mathbb{E}\, V_{t+1}(a', d'_o, y', z', c'_o) + (1 - s_t)\ \max_g B\big(b(a' - g, d'_o, c'_o),\ d'_o,\ g\big)\Big] \tag{6}$$

where $s_t$ is the survival probability, and death between periods leaves the estate a′. The net bequest is

$$b(z, d', c') = z - \tau_b \max\left\{ z + \mathbb{1}[\text{no shield}]\, d' - E_{c'},\ 0 \right\} \tag{7}$$

with $E_{c'}$ = max(E − d′, 0) for a founder, since trust funding has used part of the exemption, and E otherwise.

**Choice probabilities.** Options are chosen with logit probabilities, so that small differences in value do not produce knife-edge jumps across the state space. Values are put in money terms by λ = u′(c*)(m + D), the marginal utility at the best option without founding times total resources. Founding sits behind a planning friction κ, drawn each period from a logistic distribution with mean $\mu_f$ and scale $\nu_f$ as a share of resources:

$$p_o = \frac{e^{v_o/(\lambda\nu)}}{\sum_{o' \in \mathcal{G}(o)} e^{v_{o'}/(\lambda\nu)}}, \qquad \pi_F = \Lambda\!\left( \frac{(\bar{v}_F - \bar{v}_B)/\lambda - \mu_f}{\nu_f} \right), \qquad \mathcal{V} = (1 - \pi_F)\, \bar{v}_B + \pi_F\, \bar{v}_F \tag{8}$$

where $\mathcal{G}$(o) is the option's group (founding or not), v̄ is a group's expected value, Λ is the logistic function and ν = 0.002 is a small smoothing scale. A household founds a trust when its gain, as a share of resources, exceeds its friction draw. The value before the shock is $V_t = \mathbb{E}_\xi[\mathcal{V}]$.

**Lifetime gifts.** At the parent's age 55, before the trust choices, the parent picks a gift g from the menu with the same logit rule. A gift costs g + $\tau_g$ max(g − X − $E_r$, 0), where X = \$300k is the tax-free allowance, and adds a warm glow

$$\phi_g \left[ \left( 1 + \frac{g}{\phi_2} \right)^{1-\sigma} - 1 \right] \tag{9}$$

with $\phi_g$ calibrated.

**Inheritance.** With 1% population growth, each household has k = $1.01^{35} \approx 1.42$ children on average, so a child has a sibling with probability 2(k − 1)/k ≈ 59%. A child inherits b/n of the net estate and a trust of d′/n, where n is the number of children, and becomes an heir. A child who has already founded a trust merges the inherited one into it.

### 3.3 Firms, government and equilibrium

**Production.** A competitive corporate sector uses Cobb-Douglas technology (capital share α = 0.36, depreciation 6% a year). The interest rate and wage follow from the corporate capital-output ratio: r = α/(K/Y) − δ. Entrepreneurial wealth in states L and H is invested in a private-business sector, so household wealth funds both corporate capital $K_c$ and business capital $K_b$.

**Government.** The government collects labour, capital-income, estate, gift and trust-income taxes and pays pensions. Government consumption is the residual, so the budget balances without a tax adjusting. This keeps the experiments clean: switching trusts off changes revenue, but no tax rate moves in response.

**Stationary equilibrium.** Given prices, households solve their problem, and the joint parent–child distribution is stationary. The capital market clears:

$$\text{excess}(r) = \frac{K^{\text{supply}}(r) - K_b(r)}{K_c^{\text{demand}}(r)} - 1 = 0 \tag{10}$$

In the calibration, r is fixed at 6% (K/Y = 3) and β is chosen so the market clears. In every experiment, all preference and technology parameters are held fixed and r is re-solved. Dollar thresholds (the exemption, fees, gift limits) are held fixed in model units so they do not drift with r.

### 3.4 Calibration

Seven parameters (Table 4) are estimated jointly to match eleven moments (Table 3); everything else is fixed at statutory values or external estimates. The estimator minimises the sum of squared residuals, each scaled by a pre-set standard error:

$$\min_x \sum_{k=1}^{11} \left( \frac{m_k(x) - \hat{m}_k}{s_k} \right)^2 \tag{11}$$

using a bounded trust-region least-squares solver (SciPy `least_squares`) with finite-difference Jacobians. Targets and standard errors were recorded in a dated project log before each calibration run. The main calibration reaches a sum of squares of 16.4 (11 moments, 7 parameters). Differences below about 0.05 are numerical noise: re-solving the best point gives values from 16.39 to 16.43.

**Table 3. Calibration targets and model fit**

| Moment | Data | Source | $s_k$ | Model, main (θ 1.04) | Model, alternative (θ 1.38) |
|---|---|---|---|---|---|
| Capital market gap at r = 6% | 0 | equilibrium | 1% | +0.2% | +0.3% |
| Top 1% wealth share, DFA-measured | 32.5% | Fed DFA, Q2 2026 | 1.0 | 32.6% | 32.6% |
| Top 0.1% share, DFA-measured | 15.0% | Fed DFA, Q2 2026 | 1.0 | 16.1% | 16.0% |
| Top 10% share, DFA-measured | 68.9% | Fed DFA, Q2 2026 | 1.5 | 70.9% | 71.0% |
| Trust share of top-1% wealth | 15% | CRS R48879 (2026): 10–20% | 5 | 18.8% | 20.6% |
| Trust share of top-0.1% wealth | 40% | Galle, Gamage & Lord (2026): one-third or more | 7 | 29.1% | 28.4% |
| Estate tax / (gross − spousal), $50M+ estates | 23.8% | IRS SOI 2024, Table 1 | 3 | 26.8% | 26.7% |

| Moment | Data | Source | $s_k$ | Model, main (θ 1.04) | Model, alternative (θ 1.38) |
|---|---|---|---|---|---|
| Share of deaths owing estate tax | 0.14% | Tax Policy Center | 0.15 (log) | 0.13% | 0.13% |
| Charity / (gross − spousal), taxable estates | 18.1% | IRS SOI 2024, Table 1 | 3 | 19.1% | 19.6% |
| Wealth of 75+ / mean | 1.53 | SCF 2022 | 0.15 | 1.09 | 1.11 |
| Lifetime gifts, share of transfers | 28% | Sabelhaus (2024), SCF 2021 | 4 | 24.4% | 24.0% |

*Note:* the lifetime-gift target was entered as 27% in all calibrations; the source figures ($247.5 billion of $879.3 billion in 2021) give 28.1%. The 1.1-point difference is about a quarter of the target's standard error.

**Table 4. Calibrated parameters**

| Parameter | Role | Main (free) | Alternative (free, earlier run) | θ fixed at 1 |
|---|---|---|---|---|
| β | annual discount factor | 0.948 | 0.947 | 0.948 |
| $\varphi_1$ | bequest strength | −1.36 | −1.55 | −1.20 |
| χ | charity strength | 0.140 | 0.149 | 0.126 |
| $\mu_f$ | founding friction, mean (share of wealth) | 0.046 | 0.075 | 0.051 |
| $\nu_f$ | founding friction, logistic scale | 0.019 | 0.026 | 0.023 |
| θ | founder's weight on trust assets | 1.04 | 1.38 | 1 |
| $\phi_g$ | gift warm-glow strength | −1.77 | −1.79 | −1.79 |
| | Sum of squares | 16.4 | 17.0 | 16.6 |

Two misses stand out. The model's top 0.1% share is 1.1 pp high and its trust share of top-0.1% wealth is 11 pp low; the two pull against each other, since more trust wealth at the very top would raise the measured share further. The old-age wealth ratio (1.09 against 1.53) is a known structural limitation: the model has no housing, spouses or late-life medical spending, and a version with medical costs did not close the gap. Its standard error was widened to 0.15 so this one moment does not drive the fit. The estate-tax targets need care too. They come from 2024 returns, filed under a lower per-person exemption than the 2026 law the model applies, and they describe individual decedents, many with a surviving spouse, whereas a model estate is a family's at the second death. We target ratios (tax and charity relative to the gross estate net of spousal transfers) rather than levels for this reason, but the mapping is approximate.

**Trust wealth in dollars.** Model trust assets are 7.2% of household wealth, about $13 trillion at Q2 2026 values, of which about 90% belongs to the top 1%. This is above the published dollar estimates ($4.5 trillion or more in Galle, Gamage and Lord 2026; $5.4–10.8 trillion in CRS 2026). Those estimates are lower bounds and

measure narrower concepts: the CRS figure covers non-grantor trusts only and is inferred from trust income, and Galle, Gamage and Lord report their most conservative estimate. Our calibration targets the share of top wealth held in trusts, where the model sits above the CRS centre for the top 1% and below Galle, Gamage and Lord for the top 0.1%. Section 4.4 shows the results across the plausible range.

**Untargeted moments.** The model gives a parent–child wealth elasticity of 0.28 and a rank-rank slope of 0.43, and 13% of top-1% households hold a trust. Personal (non-trust) wealth puts 30.6% in the top 1%; total wealth, trusts included, puts 34.5% there.

**Choice of θ.** The targets pin down θ only weakly, so we report how the main calibration was reached. A first run with all seven parameters free stopped at θ = 1.38 (sum of squares 17.0); it is the alternative calibration and gives a smaller effect (+2.4 pp). Fixing θ = 1 and re-estimating the other six parameters fitted better (16.6). Freeing θ again from that point, the estimator settled at θ = 1.04 with the best fit (16.4): raising θ further worsened the fit, and its trial steps returned to that point. Values well below one were not explored, since the search moved up from one. An earlier version (v2b) matched the top shares on personal wealth alone; its larger effect (+3.7 pp) comes mostly from its θ of 0.94, not from the change of target. §3.5 explains why the DFA-consistent measure is the right target.

### 3.5 Measuring top shares the way the data do

A model with trusts has to say which wealth the data count, and for whom. We match model and data by measuring model wealth the way the Federal Reserve's Distributional Financial Accounts (DFA) measure it.

Three facts drive the construction:

1. **Trust assets are in the DFA totals.** The household sector of the Financial Accounts, which the DFA distributes, explicitly includes personal trusts (Fed Z.1 table guide).
2. **The DFA spreads totals by asset class.** Each Financial Accounts total is allocated to households in proportion to their holdings of that asset class in the Survey of Consumer Finances (Batty et al.).
3. **The SCF records few trusts.** Trust assets the SCF records come to 2.31% of net worth (SCF 2022); in the model, trusts hold a share τ of total wealth that is several times larger.

So in the DFA, a dynasty trust's assets are counted, but mostly not attributed to the family that benefits. They are spread across all holders of the same assets, mainly corporate equity, of which the top 1% hold 50.9% (DFA Q2 2026). We replicate this. A household's measured wealth is

$$x_i = a_i \;+\; c\,d_i \;+\; (1-c)\,D\,\frac{a_i^{\eta}}{\sum_j a_j^{\eta}}, \qquad c = \min\!\left(1, \frac{0.0231}{\tau}\right) \tag{12}$$

where $a_i$ is personal wealth, $d_i$ trust assets the household benefits from and D total trust assets. The first term is wealth the SCF sees directly. The second is the share of trusts the SCF records, attributed to the right family. The third spreads the rest by holdings of equity-like assets; η is set so that households in the top 1% of personal wealth receive 50.9% of it.

**Why it matters.** The measure sits between two bounds: personal wealth (trusts invisible) and total wealth

(trusts fully attributed). In the main calibration the top 1% hold 30.6% of personal wealth, 32.6% measured, and 34.5% of total wealth. Targeting the measured share stops the calibration from either ignoring trust wealth or counting it where the data cannot see it. It also yields a second result: the effect of trusts that official statistics would record is smaller than the true effect (§4).

**Sensitivity.** At the earlier v2b parameters, the measured top-1% share ranges from 33.0% to 35.0% across an equity share for spreading of 0.45 to 0.55 and an SCF capture rate c from 0 to 0.5.

### 3.6 Experiments

Each experiment (Table 5) changes one feature of the calibrated economy, holds every preference and technology parameter fixed, and re-solves the interest rate. We compare stationary equilibria, so results describe the long run, after the economy has fully adjusted.

**Table 5. Experiments**

| Experiment | Change from the calibrated economy |
|---|---|
| base | none: dynasty trusts (shield and lock-in) plus lifetime gifts |
| off | no dynasty trusts; lifetime gifts still available |
| lock0 | tax shield kept, lock-in removed (heirs can reach all principal) |
| noshield | lock-in kept, tax shield removed (trust assets enter the taxable estate) |

**Decomposition.** The total effect of trusts on a top share S is S(base) − S(off). It splits into the effect of each feature alone and their interaction:

$$\underbrace{S_{\text{base}} - S_{\text{off}}}_{\text{total}} = \underbrace{(S_{\text{lock0}} - S_{\text{off}})}_{\text{shield alone}} + \underbrace{(S_{\text{noshield}} - S_{\text{off}})}_{\text{lock-in alone}} + \text{interaction} \tag{13}$$

**True and measured effects.** Each effect is reported twice: on total wealth (trusts attributed to their beneficiaries) and on the DFA-consistent measure of §3.5, which is what official statistics would show.

**Partial vs general equilibrium.** The first evaluation of every experiment holds r at 6% (partial equilibrium). The reported results re-solve r (general equilibrium).

**Trust-share sweep.** The trust share of top-1% wealth is the least certain target (10–20% in CRS; one-third or more of top-0.1% wealth in Galle, Gamage & Lord). To show how the answer depends on it, we re-solve the founding-friction mean $\mu_f$ and β jointly so that the trust share hits 10%, 15% and 27% while the capital market still clears at 6%, with all other parameters at the alternative calibration (θ = 1.38). Then we repeat the base vs off comparison at each point. These targets and the procedure were recorded in the project log before any sweep run.

## 4. Results

All results compare stationary general equilibria: one experiment changes, every preference and technology parameter stays at the calibrated value, and the interest rate is re-solved. The main experiments converged with a capital-market gap of 0.04% or less, and every robustness point below 0.2%.

### 4.1 The effect of dynasty trusts

Removing dynasty trusts lowers the top 1% share of total wealth from 34.5% to 31.3% (Table 6), a fall of 3.2 percentage points. The top 0.1% share falls by 3.1 points (18.1% to 14.9%) and the top 10% share by 1.8 points. Calibrations that fit the data almost as well give a top-1% effect of 2.4 points ($\theta = 1.38$) and 3.6 points ($\theta = 1$).

**Table 6. Top wealth shares by experiment (general equilibrium)**

| Experiment | Top 1% | Top 1%, DFA-measured | Top 0.1% | Top 10% | Trust share of top-1% wealth | Estate tax, % of output | Interest rate |
|---|---|---|---|---|---|---|---|
| Dynasty trusts (shield + lock-in) | 34.5% | 32.6% | 18.1% | 71.6% | 18.8% | 0.15% | 5.99% |
| Funding capped at the GST exemption | 32.7% | 32.1% | 16.1% | 70.7% | 10.7% | 0.21% | 6.09% |
| Shield only, partial withdrawals | 31.8% | 31.8% | 15.5% | 70.2% | 4.2% | 0.17% | 6.18% |
| Shield only: heirs can reach all principal | 31.7% | 31.7% | 15.3% | 70.1% | 2.5% | 0.18% | 6.19% |
| Lock-in only: trust taxed at death | 31.6% | 31.5% | 15.0% | 70.1% | 5.3% | 0.27% | 6.15% |
| No dynasty trusts | 31.3% | 31.3% | 14.9% | 69.9% | — | 0.25% | 6.21% |

*Notes: top shares of total wealth, with trust assets attributed to their beneficiaries, except the DFA-measured column (Section 3.5). Main calibration ($\theta = 1.04$). In the alternative calibration ($\theta = 1.38$) the top-1% effect is 2.4 points, with shield alone +0.1 and lock-in alone +0.0.*

Trusts add to saving rather than relabel existing wealth. At the calibrated interest rate, household capital is 6.0% lower without trusts. In equilibrium, the extra capital lowers the interest rate by 21 basis points. Broad mobility changes little: at fixed prices, the parent–child wealth elasticity is 0.283 with and without trusts, and the rank-rank slope is 0.43 and 0.44. In general equilibrium the elasticity without trusts is higher (0.299), because the higher interest rate raises the return on inherited wealth. Trusts concentrate the top tail without making the whole distribution more persistent.

Trusts also cut estate-tax revenue. Estate tax falls from 0.25% of output without trusts to 0.15% with them, while gift tax rises slightly (0.09% to 0.12%). For families dying with more than $30 million, estate tax as a

share of everything they pass on, trust assets included, falls from 26% to 11%. Because government consumption is the residual, the net budget effect is smaller (16.04% to 16.00% of output), partly because the larger capital stock yields more capital-income tax.

### 4.2 Mechanism: the shield and lock-in work together

The two features of a dynasty trust do little on their own: their interaction accounts for four-fifths of the effect.

- **Shield without lock-in.** If heirs can reach the principal, trusts still form, but heirs draw them down. Trust wealth falls to 2.5% of top-1% wealth and the top-1% share is 0.38 points above the no-trust economy. Letting heirs take 10%, 25% or 50% of the principal instead of all or nothing raises this only to 0.53 points (4.2% of top-1% wealth).
- **Lock-in without shield.** If trust assets are taxed at each death, many families still found trusts (10% of top-1% households, holding 5.3% of top-1% wealth), but mostly to hold wealth that would otherwise be held directly. The top-1% share is 0.25 points above the no-trust economy.

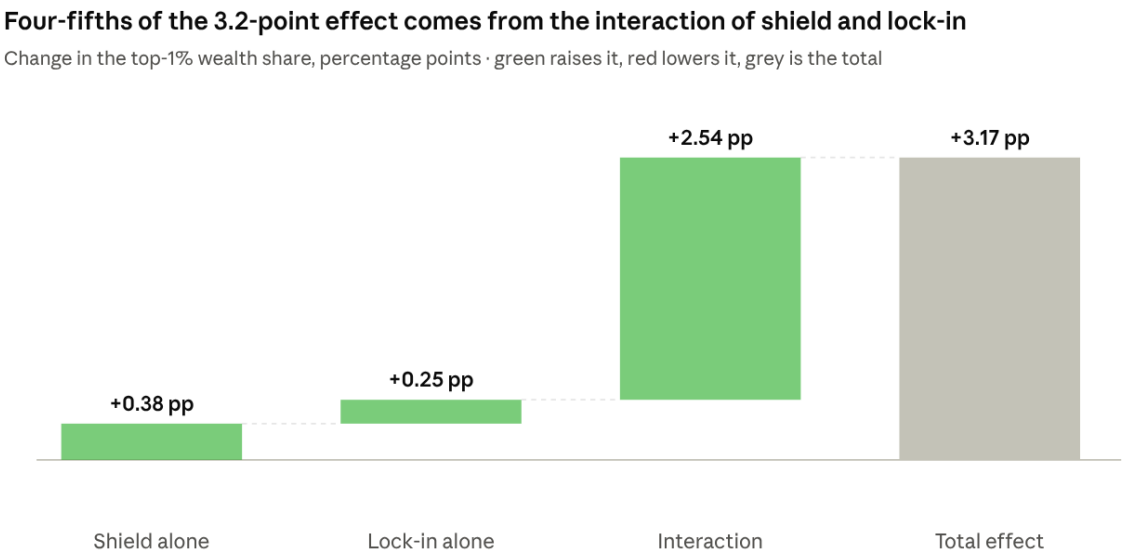



Figure 1. Decomposition of the effect of dynasty trusts on the top-1% wealth share. Model results, main calibration ($\theta = 1.04$), general equilibrium.

The interaction is the economics of the dynasty trust. The tax shield makes it worth keeping wealth in the family across generations, and lock-in stops each generation from consuming it. Policy acts on this interaction: periodic taxation of trust assets removes the shield, and a duration limit removes both features once each trust ends (Section 5.1).

The shield-only experiment lets heirs withdraw either all principal or none, not part of it. This likely overstates how fast heirs draw trusts down, but allowing partial withdrawals changes little (above). The experiment is a decomposition device rather than a legal design: a beneficiary who can withdraw all principal holds a general power of appointment, which brings the assets into the beneficiary's estate (IRC §2041).

### 4.3 What official statistics would show

Measured as the DFA would measure it (by official statistics we mean the DFA throughout), the top 1% share is 32.6% with trusts and 31.3% without, a difference of 1.28 points, against a true effect of 3.17. The level is understated too: with trust assets attributed to their beneficiaries, the top 1% hold 34.5% of wealth, about 2 points more than the DFA-measured 32.6% to which the model is calibrated. For the top 0.1% the measured effect is 1.17 points. The rest of the effect is still in the measured totals, but spread across all holders of equity-like assets rather than attributed to the families who benefit. Official top shares would therefore understate both the level of concentration and its rise when trusts spread. How large the hidden part is depends on two measurement inputs we cannot pin down precisely: the share of trust assets the SCF records and attributes to the right family, and how the DFA spreads the rest. At the DFA's own spreading share (the top 1% hold 50.9% of equities), the measured effect is 0.6 to 1.7 points as the SCF's capture rate runs from zero to one-half: between 45% and 80% of the effect is hidden. The spreading share matters more. If the hidden trust wealth is spread so that the top 1% receive 45% to 65% of it, the measured effect at our capture rate runs from 1.0 to 2.0 points. The alternative calibration gives the same picture at a lower level (0.76 points measured at the main measurement assumptions). Allocating trust wealth by beneficiary would remove this uncertainty.

### 4.4 How much depends on the size of trust wealth

The least certain input is the share of top wealth held in trusts. This sweep was run on the alternative calibration ($\theta = 1.38$). The main calibration gives a larger effect at a lower trust share (+3.2 points at 18.8%), so the sweep probably understates the effect at each trust share. We re-solve the founding-friction mean and the discount factor so that trusts hold 10%, 15% and 27% of top-1% wealth, with the capital market still clearing at 6%, and repeat the comparison at each point.

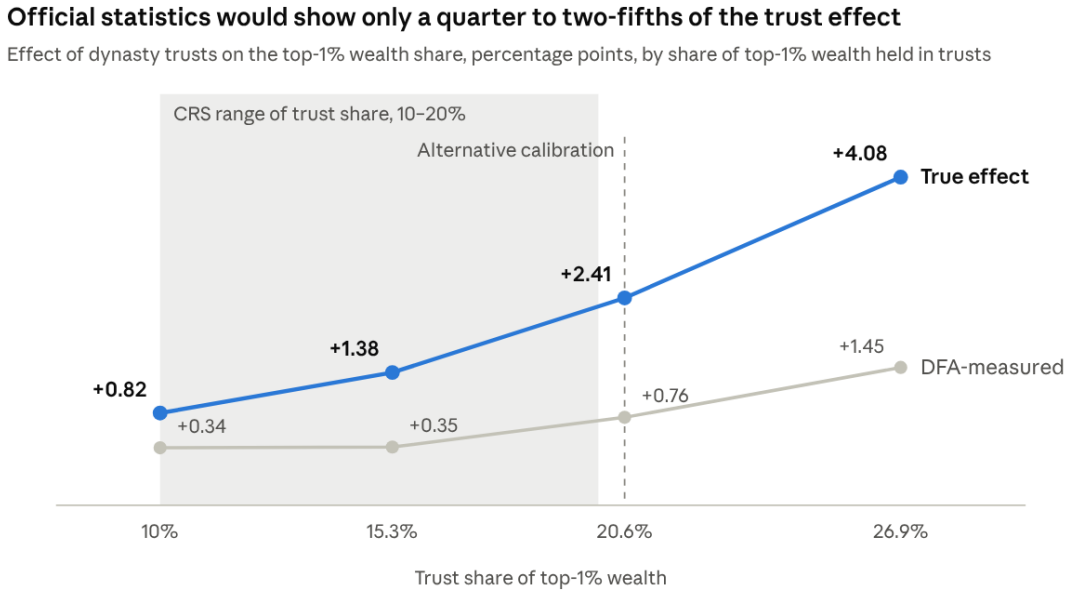



Figure 2. Effect of dynasty trusts on the top-1% share by the share of top-1% wealth held in trusts, true and DFA-measured. Trust-share sweep on the alternative calibration ($\theta = 1.38$); CRS range from CRS R48879 (2026).

The effect rises faster than the trust share, from about 0.11 points per percentage point of trust share at the low end to 0.27 at the high end. At 15%, the centre of the CRS range, the effect is +1.4 points. Because the sweep re-solves only the founding friction and the discount factor, personal-wealth concentration barely moves: the top-1% share of personal wealth stays between 30.6% and 30.9%. A higher trust share therefore means more total wealth at the top rather than a relabelling of the same wealth, which is why the effect rises faster than the trust share. The DFA-measured effect is between a quarter and two-fifths of the true effect, and about a third at that calibration's own trust share. The ratio is not monotone in the trust share, probably because two forces offset: the recorded share $c = 0.0231/\tau$ falls as trusts grow, while the spreading rule hands more of the unattributed wealth to the top. Near the centre of the Galle, Gamage and Lord (2026) estimate for the top 0.1% (about 38% trust share there), the top-1% effect is about 4 points.

### 4.5 Robustness

- **Calibration target.** A calibration that matches top shares on personal wealth alone (v2b) gives a top-1% effect of 3.7 points, with a similar decomposition. Most of the gap comes from the founder's valuation of trusts ($\theta = 0.94$ in v2b, against 1.04 and 1.38 in our DFA-consistent calibrations) rather than from the change of target: with $\theta = 1$ at our targets the effect is +3.6.
- **Grid.** At the v2b parameters, raising the trust grid from 55 to 70 points moves the effect from 3.67 to 3.57 points.
- **Partial vs general equilibrium.** Holding the interest rate at 6% changes the top-1% effect by 0.1 points or less.
- **Founder's motive.** Calibrations that fit almost as well give +2.4 points ($\theta = 1.38$) and +3.6 points ($\theta = 1$), against +3.2 in the main calibration. Adding freeze transfers into the trust at $\theta = 1$ changes nothing (Section 5.2).
- **Lifetime gifts and the exemption.** The model lets lifetime gifts above the allowance use the exemption without reducing it later, which makes large gifts too cheap. At the opposite extreme, where such gifts get no exemption, the top-1% effect in the alternative calibration rises from +2.4 to +2.9 points (+0.9 DFA-measured). With gifts expensive, families substitute into trusts (trust share of top-1% wealth 22.5%), and the economy without trusts loses an avoidance route. The true rule lies between the two.
- **Return on trust assets.** Professional management, a long horizon and retained founder positions could let trust assets earn more than the same assets held directly. We check this in the alternative calibration (effect +2.4). With an extra 0.5% or 1% a year on trust assets, and the founding friction re-solved so that trusts still hold about 20.6–20.8% of top-1% wealth, the top-1% effect is +2.5 points in both cases. Holding all other parameters fixed instead, a 1% premium raises the trust share to 24% and the effect to +3.3 points at a 6% interest rate. A return advantage therefore matters only through how much trust wealth accumulates, which the calibration pins to outside estimates.
- **Funding above the GST exemption.** The baseline lets a founder pay gift tax on funding above the exemption and then shield all of it; in law that excess would face GST tax at each generation. Denying the shield to such funding altogether cuts trust wealth to 10.7% of top-1% wealth and the effect to +1.4 points (+0.8 DFA-measured). That is close to what the trust-share range implies at the same trust share (Figure 2), so the rule works mainly by keeping wealth out of trusts. Since the calibration pins trust wealth to outside estimates, this point is best read as one more point on that range. In practice families

grow trusts beyond the exemption with freeze transfers instead.

- **Partial withdrawals.** In the shield-only experiment, letting heirs take part of the principal raises the effect from +0.38 to +0.53 points; lock-in remains necessary.

## 5. Discussion

### 5.1 Policy

The decomposition shows which features of a trust a reform must reach, but neither single-feature experiment is itself a policy: in law the two features come together, because a trust whose beneficiaries can withdraw all principal is in their estates (Section 4.2). The legal levers act as follows.

- **Duration limits.** Restoring a rule against perpetuities, or a federal limit on how long GST-exempt status lasts, ends each trust or its exemption after a fixed horizon, typically 90 years or lives in being plus 21. The assets then pass to beneficiaries outright or become subject to GST tax, losing both features, but only once every two to three generations. Our steady-state experiments do not capture such a horizon; an experiment in which trusts lose their status after a fixed number of generations is the natural next step.
- **Periodic taxation of trust wealth.** Taxing trust assets once a generation, as the estate tax would if they passed through an estate, removes the shield while leaving trusts in place. This is our lock-in-only experiment: the effect then nearly disappears (+0.25 points), and trusts become a way to hold wealth rather than to accumulate it.
- **The GST exemption and distribution rules.** The GST exemption, now $15 million per person, sets how much can enter a trust free of transfer tax. The trust-share range in Section 4.4 shows how the effect scales with trust wealth, which a lower exemption would reduce. It is only a guide, because a lower exemption would also raise the estate tax that trusts avoid. Rules requiring trusts to distribute a minimum share of their assets each year would weaken lock-in directly. Our checks suggest that weakening lock-in removes most of the effect even when heirs draw trusts down gradually (Section 4.5), but we do not model such rules.

Both comparisons are between steady states. They describe where the economy ends up, not the path there, and they leave out avoidance responses outside the model, such as moving assets offshore. A reform that applies only to new trusts would also take generations to have its full effect, because existing perpetual trusts would keep their status.

Measurement policy matters too. Because the DFA spreads most trust wealth across all equity holders, official top shares understate the level of concentration and would understate its rise if trust wealth grows. Better reporting of trust assets by beneficiary, for example from fiduciary returns linked to beneficiaries, would let official statistics see what the model shows.

### 5.2 The founder's valuation of trust assets

The main calibration gives founders a weight of $\theta = 1.04$ on trust assets relative to direct bequests: a dollar left in trust counts about as much as a dollar given outright, so founders value trusts mainly for what they do to taxes and heirs' access. Several features could still justify a value above one. Trusts give founders control over how heirs use the money and protect it from heirs' creditors, divorces and spending, which the estate-planning

literature cites as main motives. The family-office literature describes these vehicles as governance machinery (spending rules, control over investment and succession) that families value in their own right (Glucksberg and Burrows 2016; Rivo-López et al. 2017, 2021; Hayoz, Ge and De Massis 2025). And our trusts are simpler than real ones: the advanced tools that make trusts more tax-efficient (valuation discounts, grantor retained annuity trusts and sales to grantor trusts) are switched off, and θ likely absorbs part of their value.

θ matters for the headline, and our targets pin it down only weakly. Three calibrations that differ mainly in θ fit almost equally well (sums of squares 16.4 at θ = 1.04, 16.6 at θ = 1 and 17.0 at θ = 1.38), but they give top-1% effects of +3.2, +3.6 and +2.4 points, and DFA-measured effects of +1.3, +1.6 and +0.8. A larger θ makes trusts attractive to merely rich families, whose trusts mostly hold wealth that would have existed anyway: at θ = 1.38, 22% of top-1% households hold a trust, against 13% at θ = 1.04. With θ near one, fewer and richer families found trusts, and the same trust wealth concentrates more. The effect therefore depends on who holds trust wealth, not only on how much there is, and the range from +2.4 to +3.6 points across these calibrations is a real uncertainty, not a bound. Adding freeze transfers into the trust at θ = 1 changes nothing (+3.58 points), so the planning tools do not stand in for θ. Even with θ near one, some families found trusts in the lock-in-only experiment, where trusts carry no tax advantage, but those trusts hold little wealth (5.3% of top-1% wealth).

**What the model identifies, and what it does not.** The calibration conditions on how much wealth is held in trusts, so any mechanism that only makes trusts more attractive is absorbed by the founding friction. A higher return on trust assets is the clearest example: it leaves the effect almost unchanged at a given trust share (Section 4.5), although it raises the effect if nothing else adjusts. The model can still distinguish mechanisms that change who holds trust wealth, as the comparison between θ = 1.38 and θ = 1 shows. What it cannot do with current data is tell why families use trusts: tax motives, control, a return advantage or planning tools fit the targeted moments almost equally well. Three kinds of evidence would sharpen the answer: trust take-up and trust size across the wealth distribution, which separate motives that spread trusts from those that concentrate them; the growth of trust assets over time, which in a model with transitions would pin down returns and frictions; and income relative to assets on fiduciary tax returns, which would measure trust returns directly. Better measurement of trust wealth itself, the anchor of our calibration, would narrow the range in Section 4.4 more than any modelling refinement.

A first look at the take-up evidence is suggestive but not decisive. In the model, how trust holding rises from the 99th to the top 0.1 percentile depends on θ but hardly at all on the amount of trust wealth: take-up in the top 0.1% is 2.37 times that in the rest of the top 1% in the main calibration (θ = 1.04), 2.16 times at θ = 1 and 1.55 times at θ = 1.38 (1.53 to 1.58 across the trust-share range). In SCF 2022, counting households with $100k or more in trusts, the ratio is 2.11, and the trust share of wealth rises from the top 1% to the top 0.1% by a factor of 1.58, against 1.55 in the main calibration, 1.54 at θ = 1 and 1.38 at θ = 1.38. Both comparisons favour θ near one, as the fit does, but the SCF's small top sample leaves all three calibrations inside sampling error (bootstrap standard deviations 0.76 and 0.20), and the SCF records trusts held for beneficiaries poorly. Take-up by wealth from fiduciary returns linked to beneficiaries would settle the question.

### 5.3 Limitations

- **Old-age wealth.** The model's wealth of households aged 75 and over is 1.09 times the mean, against 1.53 in the SCF. The model has no housing and a wealth-to-earnings ratio of about 4.9 against about 7.5

in the data, so pensions dominate retirement resources. Adding late-life medical costs moved the ratio by less than 0.02, and a version with life extended to 94 did not raise it. We down-weight this target.

- **Trust wealth at the very top.** The two outside estimates imply that trust wealth is much more concentrated in the top 0.1% than the model produces (29% of top-0.1% wealth in trusts, against ≥ 33% in Galle, Gamage and Lord). In SCF 2022, though, the ratio of trust shares between the top 0.1% and top 1% is 1.6–1.7, close to the model's 1.5.
- **Steady states, closed economy.** We compare long-run equilibria and ignore transitions, offshore trusts and non-citizen channels. This matters more than usual here. Perpetual trusts are at most about 40 years old in most states, so today's trust stock is far from a steady state, and the trust-share targets describe a stock that is still building. Our effects are the long-run consequence of current law, not the effect on today's top shares.
- **No tax basis.** Capital income is taxed as it accrues, at an effective 18%, so the model has no unrealized gains, no step-up in basis at death and no carryover basis for gifts and trusts. This probably overtaxes buy-and-hold founders while they are alive, and it overstates the trust's advantage when a trust holding low-basis assets eventually sells them (Section 6).
- **Simplified trust law.** Heirs who add to an inherited trust merge it with their own, and the founder's remaining exemption is approximated by the current trust value. Both are conservative, and are listed in the computational appendix.

## 6. Conclusion

In our main calibration, dynasty trusts raise the top 1% wealth share by about 3.2 percentage points in the long run, and the top 0.1% share by 3.1 points. Calibrations that fit the data almost as well give 2.4 to 3.6 points, depending on how much founders value trusts beyond their tax advantages, and the effect scales with the amount of trust wealth, which is poorly measured. The effect comes from the combination of the tax shield and lock-in, not from either alone, so periodic taxation of trust wealth would remove most of it in the long run. Much of the effect is invisible in the statistics used to track US wealth concentration, because they count trust assets but cannot attribute them to the families who benefit.

Several extensions follow naturally. The first is the transition: how fast concentration would rise as perpetual trusts accumulate, and how fast it would fall under a reform that applies only to new trusts. The second is the advanced planning tools we largely switch off here, such as valuation discounts and freeze transfers, which change how much families can move outside the estate tax. A third question is new: recent large pledges to seed children's savings accounts offer the very wealthy a tax-advantaged way to give that spreads wealth rather than concentrating it, and the same model can compare the two.

The most important extension is to model the tax basis of assets. Our households pay tax on capital income each period as it accrues, so the model has no unrealized gains, no choice of when to sell, and no step-up in basis at death. In practice these features are central to how large fortunes are taxed. Founders hold appreciated stock for decades and pay little tax while alive, borrowing against it rather than selling, and their heirs inherit it with the gain erased. A dynasty trust changes this trade-off. Assets inherited directly pay estate tax above the exemption but receive a step-up; assets in a trust escape estate tax but keep their original basis, so the trust pays capital-gains tax whenever it sells. For low-basis founder stock, the trust's advantage per generation therefore

lies between about 16 percentage points (40% estate tax less 23.8% capital-gains tax, if the trust eventually sells) and the full 40 points (if it never does). Grantor trusts narrow the gap further, because founders can swap low-basis assets back into their own estate before death to collect the step-up. Lifetime gifts and trust funding carry over the giver's basis, so they too forgo the step-up.

Adding basis requires a further state variable for each household and its trust, a decision about when to realize gains, and the rules for step-up, carryover basis and asset swaps. That multiplies the computational burden several times over, beyond what a single desktop computer can calibrate in reasonable time, but it is feasible with parallel computation. It would also let the model address current proposals to tax unrealized gains at death or limit the step-up, and how such reforms interact with dynasty trusts. We leave this to future work. Within this paper, the multi-generation structure limits how much basis can matter: the estate tax that a trust avoids recurs at every generation, while the step-up benefit of direct inheritance applies to one generation's gains at a time.

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## Appendix A. Computation

All results come from about 3,500 lines of Python (NumPy, SciPy, Numba), run on a single personal computer (Intel Core i7-1065G7, 4 cores and 8 threads, 16 GB RAM). Every model evaluation is cached to disk, so every run can be restarted and every reported number traced to a logged evaluation.

The distribution loop (highlighted) is where trusts make the computation slow, and where the convergence test matters most for top shares (A.2).

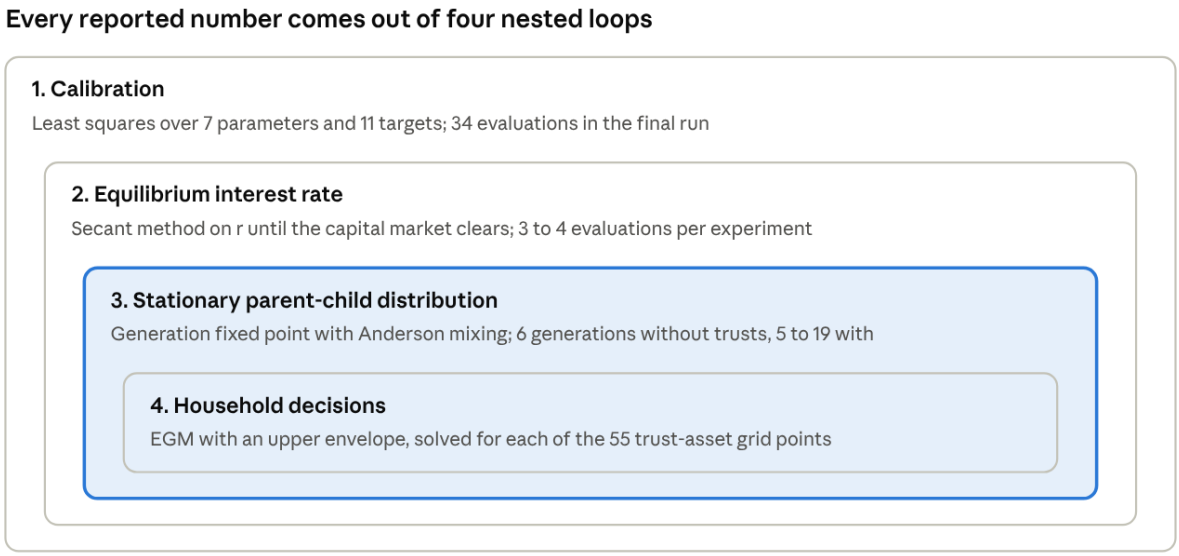



Figure A1. Computational structure: every reported number comes out of four nested loops.

### A.1 Household problem

For each trust-asset grid point and next-period household category, the liquid-saving problem is solved by the endogenous grid method (EGM), with a DC-EGM upper envelope for the non-concavities that discrete choices create. Discrete choices (founding, contribution size, gift size, retaining or paying out trust income) are smoothed by small logit shocks (scale 0.002 of wealth). Off-grid trust choices are interpolated along lines of constant total wealth. Return shocks are integrated with Gauss-Hermite quadrature. Table A1 lists the grids.

**Table A1. Grids and quadrature**

| Grid | Points | Range |
|---|---|---|
| Liquid wealth | 160 | up to $500B |
| Trust assets | 55 | from $250k |
| Earnings states | 5 | Rouwenhorst |
| Return states | 3 | N, L, H |
| Return shocks | 5 | Gauss-Hermite nodes |
| Ages | 14 | five-year periods, 20–89 |

### A.2 Stationary distribution

The distribution is computed directly on the grid, not by simulation, so top shares carry no Monte Carlo noise. A child's path is independent of the parent's until the parent dies, given the child's initial type. The parent–child joint distribution is therefore kept in factorised form, which is exact and keeps the two-generation state space tractable. This requires life to be no longer than two generations ($T \leq 2 \times \text{gap}$), which is checked in code.

The distribution is the fixed point of a map from one generation of parents to the next, accelerated with Anderson mixing. Convergence requires both a change in mass below $10^{-6}$ and a wealth-weighted change below $10^{-4}$. The wealth-weighted test was added after we found that a mass-only test stopped before slow-building trust wealth at the very top had settled; that early stop understated the trust share.

### A.3 Equilibrium and calibration

The interest rate is found by the secant method on the capital-market excess, safeguarded by bisection once a bracket is found, to a tolerance of 0.2%. All main experiments converged with |excess| ≤ 0.04%, and all robustness and sweep points within 0.2% (largest 0.16%). Each evaluation warm-starts from the previous distribution.

The calibration uses SciPy's trust-region reflective least squares with per-parameter finite-difference steps (0.1% for β, 3–5% for the others; a larger β step was found to jump across the market-clearing region). The first DFA-consistent calibration started from the previous version's optimum and was stopped at evaluation 34, when the sum of squares had flattened (the best value improved from 17.16 at evaluation 21 to 17.04 at evaluation 33, and evaluation 34 was worse). That run is the alternative calibration. The main calibration came from two further runs: one with θ fixed at 1, stopped by hand by the same rule (best 16.60), and one with all seven parameters free, started from that point and stopped automatically once the best fit had improved by less than 0.1 over 16 evaluations (26 evaluations, best 16.39 at θ = 1.04).

### A.4 Accuracy checks

Table A2 summarises the checks on the numerical solution.

**Table A2. Accuracy checks**

| Check | Result |
|---|---|
| Nesting: trusts off reproduces Layer B; no entrepreneurs reproduces Layer A | exact (test suite) |
| Trust grid 70 vs 55 points (v2b, GE) | effect +3.57 vs +3.67 pp; trust share 20.6% vs 21.3% |
| DFA measure reproduces the offline calculation at the v2b point | 34.10% / 17.13%, identical |
| DFA measure returns personal shares when there are no trusts | yes (off experiments) |
| Capital-market clearing, all experiments | main experiments ≤ 0.04%; robustness and sweep points ≤ 0.16% |

The grid check moved the headline by 0.1 pp. Going from 40 to 55 points had moved the trust share by about 9 points; going from 55 to 70 moved it by 0.6, so 55 points is adequately converged.

### A.5 Run times

Table A3 gives the computing time for the main runs, on a single desktop PC.

**Table A3. Run times**

| Run | Evaluations | Time per evaluation | Wall time |
|---|---|---|---|
| Alternative calibration (θ = 1.38) | 34 | 34–64 min | 24.5 h |
| Experiments: base, off, noshield, lock0 | 3–4 each | 2 min (no trusts) to 58 min | 6.6 h |
| Trust-share sweep, three points | 2–5 per point + off | 35–65 min | 7.5 h |

Economies without trusts converge in about 6 generations and solve in 2 minutes. With trusts, convergence takes 5–19 generations (13–17 for the calibrated economy).

The robustness runs took 10.8 hours for the two θ = 1 points, 3.3 hours for the gift-exemption bound and 4.8 hours for the two trust-return points. Two switches added for these runs, the trust-return premium and the gift-exemption rule, reproduce the main model exactly when set to their defaults (all reported moments identical). Later checks added further switches (partial withdrawals, funding capped at the exemption, the corrected lock-in-only experiment), each tested the same way. The two further calibrations took about 20 hours (θ fixed at 1) and 27 hours (θ free, 35 evaluations of which 9 were replayed from the cache), and the main-case experiments about 12 hours.

### A.6 Reproducibility

Each run is started by a one-line launcher and writes JSON results and a plain-text log. Before each calibration and experiment, the targets, standard errors and procedure were written into a dated project log (`DECISIONS.md`), which also records every change to the model, every failed variant and the reason for each choice. One variant was abandoned after running: a founding friction that falls with wealth produced trust shares of 56–80% and was discarded. The code, logs, cached evaluations and launchers are publicly available at https://github.com/nestegglab/hidden-dynasties.

### A.7 Use of AI tools

The model code, calibration and experiment scripts were written and run with the assistance of Claude, an AI model made by Anthropic, working under the author's direction. Claude also assisted in drafting and editing the text of the paper. Independent reviews of the model and draft were obtained from separate AI model sessions. All modelling choices, targets and pre-registrations were approved by the author, and the author is responsible for the results.